

K Yuvraj

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OBJECTIVE

Detail-oriented **Mechanical Engineer** with experience in **Product Lifecycle Management (PLM)**, **SolidWorks**, and **GD&T**. Proven background in **new product development** for health/hygiene solutions, familiar with documentation rigor required for compliant industries. Skilled in **component selection**, **machining processes (CNC)**, and **BOM creation**, with exposure to process equipment and reactor instrumentation.

EDUCATION

- **B.Tech in Mechanical Engineering, Institute of Aeronautical Engineering** 2024
Vice - President at E-CELL, the college entrepreneurship club
- **Diploma in Mechanical Engineering, TKR College of Engineering and Technology** 2021
Department Representative

SKILLS

- Technical Skills:** Product Design, Product Development, Project Management, Product Lifecycle Management, Additive Manufacturing, Innovation and RnD, FMEA, Python, HTML and CSS.
- Designing Tools:** AutoCAD, Fusion360, Solidworks, Blender
- Prototyping Tools:** Ansys, Catia, Prusa, Simscale
- Business Tools:** Microsoft Projects, Microsoft Teams, Google Analytics, Power BI, Tableau
- Soft Skills:** Leadership, Team Management, Communication, Time Management, Empathy, Negotiation, Persuasion, Conflict Resolution

PROFESSIONAL EXPERIENCE

- **Co-Founder, Head - Product Development**  August 2022 - Present
Start-up Hyderabad,IN
 - Lead **product development and technology innovation** for a sustainability start-up.
 - Designed and developed **mechanical components** using **Fusion 360 and SolidWorks** for product prototyping and manufacturing.
 - Oversee **daily operations, strategic planning, and team coordination** to drive business growth.
 - Utilized **Python and data analytics** to optimize inventory and track product lifecycle metrics.
- **Human Resource Intern**  October 2022 - January 2023
e-DAM Hyderabad,IN
 - **Managed recruitment, employee relations, and performance management** to support organizational growth.
 - Implemented training and development strategies to enhance **employee engagement and productivity**.
- **In-Plant Trainee**  July 2022 - July 2022
SriKrishnaa Processs Technologies Pvt Ltd Hyderabad,IN
 - Acquired **in-depth knowledge of Chemical Reactors and its operations**, including its various components and machining processes involved.
 - Used **Auto CAD and Fusion 360** predominantly for designing and synthesis of the chemical reactors.
- **CNC Trainee**  April 2020 - December 2020
National Small Industries Cooperartion Pvt Ltd Hyderabad,IN
 - Acquired **in-depth knowledge of CNC (Computer Numerical Control) and NC (Numerical Control) operations**, including programming, setup, and execution of machining processes.
 - Developed proficiency in **CAD/CAM software**, learning how to create precise **2D and 3D models** for CNC machining applications.
 - Gained hands-on experience in **toolpath generation, material selection, and optimization of machining parameters** to improve manufacturing efficiency.
 - Learned industry-standard **manufacturing processes, quality control techniques, and safety protocols** to ensure precision and compliance in machining operations.

PROJECTS

CFD Analysis on Muffler

- Conducted **Computational Fluid Dynamics (CFD) analysis** to evaluate the **flow dynamics and acoustic performance** of a muffler.
- Simulated **pressure drop, velocity distribution, and temperature effects** using ANSYS Fluent.
- Optimized the **internal geometry** to improve noise reduction and exhaust efficiency.

Cyber Scooter (e-Scooter) Design

- Designed an **electric scooter (e-Scooter)** with a focus on **ergonomics, efficiency, and sustainability**.
- Developed a **3D CAD model** using Fusion 360, incorporating **lightweight materials and modular components**.
- Integrated **battery management and motor drive systems** for improved performance and range.

Grass Cutting Machine using Copper Wire Blade

- Designed and prototyped a **low-cost, efficient grass-cutting machine** using **copper wire blades** for precision cutting.
- Created **detailed CAD models** and performed **stress analysis** to ensure structural integrity.
- Focused on **improving cutting efficiency and safety features** for better usability.

Fork Lifter Component Design for 3D Printing

- Designed **functional mechanical components** for a **fork lifter**, ensuring **strength and durability**.
- Optimized the model for **3D printing**, reducing material usage while maintaining structural integrity.
- Generated **technical drawings and 3D CAD models** for manufacturing and testing.

Mid-Air Part Transfer in Aircraft Assembly

- Developed a **robotic assembly method** for **mid-air part transfer**, eliminating reliance on intermediate surfaces.
- Designed algorithms to determine **optimal gripping points** for smooth, precise robotic transitions.
- Focused on **automation, efficiency, and precision** in **aerospace manufacturing**.

Chassis Design for a Caravan

- Designed a **structurally sound and lightweight chassis** for a **caravan**, balancing **strength, weight, and cost**.
- Utilized **rectangular tube structures and finite element analysis (FEA)** to optimize **load distribution**.
- Developed **detailed technical drawings and CAD models** for **CNC manufacturing**.

Mobile Drone Ambulance

- Designed a **conceptual mobile drone ambulance** for **rapid emergency response in inaccessible areas**.
- Developed a **3D CAD model** integrating **autonomous flight technology, payload optimization, and modular design**.
- Focused on **safety, stability, and real-time medical assistance feasibility** through innovative engineering.

HONORS AND AWARDS

- Guided the Entrepreneurship Cell to the finals of the National Entrepreneurship Challenge, securing **All India Rank 4th**.
- Secured Multiple Grants for the Start-up from reputed organizations and incubation centers.
- Achieved Best Innovation Award from IIC at the Regional Meet.

LEADERSHIP EXPERIENCE

Vice-President of Entrepreneurship Cell Guided and helped startups to develop by providing them with mentorship sessions and seminars.

PR and Operations Head of e-DAM Guided and helped students to upskill by providing them with mentorship sessions and seminars. Managed the internal operations of the community.